a Separate temporal designs

Steps: PaySim / BankSim; IEEE-CIS: % of all labelled rows.

PaySim

Mobile-money simulation

6,362,620 rows

	Train	Val	Test
F1	1–205	206–238	239–281
F2	1–281	282–323	324–354
F3	1–354	355–398	399–743

LightGBM (5 seeds)

Later logistic

IEEE-CIS

Anonymised transactions

590,540 rows

	Train	Val	Test
F1	40%	10%	10%
F2	60%	10%	10%
F3	80%	10%	10%

Test: 59,054 rows/fold

LightGBM; later logistic

BankSim

External payment simulation

594,643 rows

Train	Val	Test
0–107	108–125	126–179

LightGBM / XGBoost (5 seeds)

Freeze before fits / Test access

B1 / seed 42: Val selection

Selected settings held fixed

b Study-specific workflows

PaySim: later exact-3% contrasts.

Inputs

- Attributes
- Time group
- Entity key
- Prior events

Schematic fields

Features

- Current context
- Past history
- Entity aggregates

Strictly earlier events

Models

Original

LightGBM; XGBoost*

Later logistic

*BankSim only

Decisions

PaySim

Val alert cap ≤3%

Threshold → Test

Test workload varies

IEEE-CIS / BankSim

Completed Test ranking

k = ceil(0.03N)

Readouts

AP by stage

Recall / capture

Test workload

Analytical cost

c Later analyses and scope

S21 | Existing outputs

Headroom / exchanges / time structure

No new fits / scores / bootstrap draws

S22 | Retrospective development

Exposed BankSim; 6 LightGBM C07 fits

Seed 42; no independent validation

Icons and split cells are schematic; studies are not pooled.